

Graph Neural Network Model Transferability Estimation via Decomposition-Augmented Discriminant Analysis

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Abstract

Model transferability estimation is a task-adaptive pre-trained model selection problem, aiming to determine the optimal model for target dataset from a model hub pre-trained on source dataset without fine-tuning. Although existing model transferability evaluation methods have made some progress, they mainly focus on image or text data in CV and NLP. In contrast, the graph structural data with GNNs models is still underexplored, due to the complexity of the graph structure and the limitations of the generalization ability of GNN models under distribution shift. To fill this blank, we first propose a Graph Neural Network Model Transferability Estimation method via decomposition-augmented discriminant analysis, named **GNNMTE**, to evaluate the transferability of GNN models on target graph dataset without fine-tuning. It only calculates the GNNMTE score to determine whether it can be effectively transferred to target graph dataset and better select the optimal model for the target graph dataset. Specifically, our proposed GNNMTE contains three core components: (1) **Dual-block SVD fusion** for obtaining the corresponding principal component information; (2) **Adaptive weighting by singular value ratio** for guiding the extraction of important principal component information on graph data; (3) **Graph discriminant analysis** for finding the optimal projection direction that separates the classes of graph data. Extensive experimental results on cross-domain graph datasets achieve excellent results, demonstrating powerful superiority.

CCS Concepts

• Computing methodologies → Machine learning.

Keywords

Graph Neural Network; Model Transferability Estimation; Graph Distribution Shift

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1 Introduction

With the rapid development of deep learning technology, model transferability estimation has gradually become a research hotspot in fields such as computer vision (CV) [14, 16] and natural language processing (NLP) [2, 7]. However, research on graph structural data and graph neural networks (GNNs) still remains underexplored, due to the complexity of the graph structure and the limitations of the generalization ability of GNN models under distribution shift [21, 46, 49]. GNN model transferability estimation is an emerging problem, which aims to evaluate and select the most suitable GNN model for a target graph dataset from a pre-trained GNN model hub that learned on source graphs. By computing transferability scores on the target dataset, the approach enables assessing model transferability without fine-tuning, thereby facilitating more efficient and effective deployment. For **relation to Web-graphs**, this study investigates the transferability of GNN models on web-based graphs, with the goal of identifying the most suitable model for each target graph and supporting a range of real-world web applications. In particular, it emphasizes bridging graph learning research with practical applications on the World Wide Web, thereby facilitating more effective and efficient deployment of GNNs.

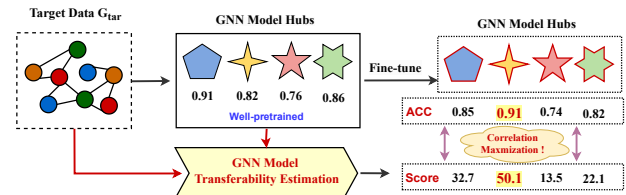


Figure 1: GNN Model Transferability Estimation.

As shown in Fig. 1, given a hub of well-trained GNN models on source graphs and a target graph, the goal is to identify which pre-trained model best suits the target graph learning and can achieve



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strong performance. Typically, a small portion of labeled nodes in the target graph is used to fine-tune these models so that they align with the new distribution, and the resulting performance is then evaluated on the unlabeled part of the target graph. However, this fine-tuning process is computationally expensive. To address this, GNN model transferability estimation leverages the labeled portion of the target graph to directly compute a score for each model in the pre-trained hub, to ensure that this score highly correlates with the eventual fine-tuned performance. This process helps improve the efficiency and effectiveness of GNN deployment in real-world scenarios. It reduces the need for costly fine-tuning and accelerates the adaptation of pre-trained models to diverse application domains.

Despite its practicality, transferability estimation in the graph domain with GNNs remains underexplored and faces two core challenges: (C1) **the inherent complexity of modeling graph structures**, and (C2) **the effective label utilization for transferability estimation**. Specifically, first, existing model transferability estimation methods only focus on features from unstructured data, while GNNs take graph-structured data as inputs, where effective transferability estimation must explicitly capture and model the complex structural relationships among nodes and edges. Second, since the calculation of the transferability estimation score should be done without fine-tuning and without using the labels of the test set of the target graph, the score is calculated solely based on the training set and labels of the target graph. However, the training subset often experiences distribution shifts compared to the test subset, which further increases the difficulty of GNN model transferability estimation.

To address these challenges, to the best of our knowledge, we are the first to propose a **Graph Neural Network Model Transferability Estimation** method via decomposition-augmented discriminant analysis, dubbed as **GNNMTE**. Through capturing the key factors that reflects whether the target graph can align with the knowledge learned by the source model, our proposed approach contains three core components: (1) **Dual-block SVD fusion**, we perform Singular Value Decomposition (SVD) dual-block fusion on node representations and structural information separately to obtain the corresponding principal component information; (2) **Adaptive weighting by singular value ratio**, we weight the components based on the ratios of singular values in different columns to guide the important information. By calculating the ratios of singular values in different columns, we can identify the components that have the greatest impact on the graph, thereby allocating more attention and weight to these components; (3) **Graph discriminant analysis**, we utilize the intra-class scatter matrix and the inter-class scatter matrix to find the optimal projection direction, designing the GNNMTE score using the graph linear discriminant analysis method to achieve maximum separation between different classes. Hence, our proposed GNNMTE leverages node and structural principal components extracted via decomposition-augmented analysis to represent the dominant patterns of the graph, applies singular value-based weighting to emphasize the most informative components, and employs discriminant analysis to measure class separability. Together, these three steps provide a principled way to approximate how well a pre-trained GNN can adapt to a target graph. In summary, our key contributions are as follows:

- **Problem.** We study a new problem, GNN model transferability estimation, which computes a GNNMTE score to evaluate the transferability of pre-trained GNN models on target graphs. This score enables the selection of the most suitable model for a given target dataset without the need for costly fine-tuning.
- **Method.** We propose a decomposition-augmented discriminant analysis framework for GNN model transferability estimation, named **GNNMTE**, with (1) dual-block SVD fusion, (2) adaptive weighting by singular value ratio, and (3) graph discriminant analysis, to effectively estimate how well a pre-trained GNN can adapt to a target graph.
- **Experiments.** Extensive experiments on real-world graph datasets demonstrate strong correlations between our GNNMTE score and the fine-tuned ground-truth node classification accuracy, showcasing its effectiveness.

2 Preliminaries

Problem definition. Let $G = (G_{\text{tar}}, G_{\text{val}}, G_{\text{te}})$ denote the target graph, where G_{tar} is the training set of the target graph, G_{val} is the validation set of the target graph, and G_{te} is the test set of the target graph which is unlabeled. We have a fully-observed training graph $G_{\text{tar}} = (X_{\text{tar}}, A_{\text{tar}}, Y_{\text{tar}})$ with the number of M nodes with C -classes of node labels, where $X_{\text{tar}} \in \mathbb{R}^{M \times d}$ is feature matrix, $A_{\text{tar}} \in \mathbb{R}^{M \times M}$ is the adjacency matrix, and $Y_{\text{tar}} \in \mathbb{R}^{M \times C}$. $Z_{\text{tar}} \in \mathbb{R}^{M \times d_1}$ is the output node embedding of training graph G_{tar} from ϕ_l .

$$Z_{\text{tar}} = \phi_l(X_{\text{tar}}, A_{\text{tar}}). \quad (1)$$

Using transferability estimation, given by an estimated score S , our goal is to rank these pre-trained models based on their performance on a given target dataset. We assess the transferability of each pre-trained GNN model using a specific metric S that produces a numerical score, denoted as T_l . Essentially, a higher score for s_l implies that the model ϕ_l is more likely to perform effectively on the given target dataset. The fine-tuning performance of each model on the target graph is denoted as $Ground_l$.

Relevance Evaluation Metric. In this paper, we use the weighted Kendall correlation coefficient τ_w as the correlation evaluation metric for the GNN model transferability estimation method. Kendall's rank correlation coefficient [8] is used to measure the degree of correlation between two ordered variables and is insensitive to outliers. The weighted Kendall rank correlation coefficient [36] takes into account the degree of influence of different migration scores on accuracy and weights the Kendall rank correlation coefficient to make it pay more attention to the models with good performance. The weighted Kendall correlation coefficient is a variant of the Kendall correlation coefficient, and the formula is as follows.

$$\tau_w = \frac{\sum_{i < j} w_{ij} \cdot (\sigma(s_i - s_j) \cdot \sigma(a_i - a_j))}{\sqrt{\sum_{i < j} w_{ij} \cdot (\sigma(s_i - s_j))^2 \cdot \sum_{i < j} w_{ij} \cdot (\sigma(a_i - a_j))^2}} \quad (2)$$

Among them, is the weight assigned to the transferability score and the precision pairs (i, j) . The range of τ_w is $[-1, 1]$. For the transferability score and the precision pairs $(T_l, Ground_l)$, $\tau_w = 1$ means T_l and $Ground_l$ are strong positive correlation ($Ground_i > Ground_j \iff T_i > T_j$), $\tau_w = -1$ means T_l and $Ground_l$ are negative correlation ($Ground_i > Ground_j \iff T_i < T_j$), and $\tau_w = 0$

means T_l and $Ground_l$ are uncorrelated. An effective method for evaluating model mobility is to generate a coefficient close to 1. The closer the coefficient is to 1, the more the designed transferability score aligns with the actual accuracy rate after transfer learning, which also indicates that our proposed method for evaluating the transferability of GNN models is more effective.

3 Our Proposed Method

As shown in Fig. 2, the proposed overall framework of GNN model transferability estimation method GNNMTE contains three stages: (1) **Dual-block SVD fusion**: the SVD process is respectively carried out on the node representation and the graph structure to obtain the corresponding principal component information. This reveals which linear combinations in the feature dimensions best explain the direction of variance, as well as the similarities and differences between nodes along the main axes of variation, determining which feature combinations or structural frequencies play a major role; (2) **Adaptive weighting by singular value ratio**: by calculating the proportion of singular values for each column relative to the total, a weight can be assigned to each column. These weights are then applied to their respective columns to guide the extraction of important information. As a result, columns with larger singular values will exert a greater influence on the final outcome, thereby emphasizing their significance and representativeness in the data; (3) **Graph discriminant analysis**: the GNNMTE score is designed by integrating a graph linear discriminant analysis approach. The intra-class and inter-class divergence matrices are utilized to find the optimal projection direction, thereby achieving the maximum separation of different classes.

3.1 Dual-Block SVD Fusion

We employ SVD to conduct an in-depth study on the output node representations and the graph structure. The SVD decomposition retains the main signal and discards the noise direction, thereby obtaining the principal component information. The consistency of the main signals between the connected nodes is maintained in the feature space and structure, and the noise directions with topological inconsistency are discarded, thereby obtaining this principal component information. In our method, the SVD on features projects nodes into the most informative subspace, reducing overfitting and computation, while providing stable input for graph linear discrimination. For graph structure, the SVD incorporates patterns to assess the alignment of the target graph with the source model, enhancing transferability evaluation. We separately perform the SVD process on the node representations Z_{tar}^l and the graph structure A_{tar} to obtain the corresponding principal component information, where $Z_{tar}^l = \phi_l(G_{tar})$ and for simplicity, we omit the subscript l for each pre-trained model.

Specifically, we first standardize the feature representations Z_{tar} and the adjacency matrix A_{tar} by column using *zscore* to make them mutually irreplaceable and indispensable components, so we have:

$$Z'_{tar} = \frac{Z_{tar}^{(i,j)} - \hat{\mu}_j}{\hat{\sigma}_j + \varepsilon}, \quad A'_{tar} = \frac{A_{tar}^{(i,j)} - \hat{\mu}'_j}{\hat{\sigma}'_j + \varepsilon}, \quad (3)$$

where $Z'_{tar} \in \mathbb{R}^{M \times d_1}$, M is the target test dataset size and d_1 is the feature dimension. The adjacency matrix after *zscore* is denoted as $A'_{tar} \in \mathbb{R}^{M \times M}$. After the *zscore* standardization operation, we apply SVD to extracted Z'_{tar} and A'_{tar} as follows:

$$Z'_{tar} = U_z \Sigma_z V_z^T, \quad A'_{tar} = U_a \Sigma_a V_a^T, \quad (4)$$

where $U_z \in \mathbb{R}^{M \times d_1}$ and $U_a \in \mathbb{R}^{M \times M}$ are orthogonal matrices of left singular vectors, while $V_z \in \mathbb{R}^{d_1 \times d_1}$ and $V_a \in \mathbb{R}^{M \times M}$ are orthogonal matrices of right singular vectors, $\Sigma_z \in \mathbb{R}^{d_1 \times d_1}$ and $\Sigma_a \in \mathbb{R}^{M \times M}$ are diagonal matrices of non-negative singular values arranged in descending order. Meanwhile, in order to retain the vast majority of effective variance information while discarding the low-energy components at the tail, it is necessary to adaptively select the k dimension. We select the number of principal components using the cumulative interpretation variance ratio, which is defined as follows:

$$k = \min\{k_0 \in \{1, 2, 3, \dots, d_1\} \mid \frac{\sum_{i=1}^{k_0} \sigma_i^2}{\sum_{i=1}^{d_1} \sigma_i^2} \geq \text{energy}\} \quad (5)$$

$$k' = \min\{k'_0 \in \{1, 2, 3, \dots, M\} \mid \frac{\sum_{i=1}^{k'_0} \sigma_i^2}{\sum_{i=1}^M \sigma_i^2} \geq \text{energy}\} \quad (6)$$

where σ_i is the i -th singular value in Σ_z and $C(k_0)$ represents the variance ratio of the cumulative explanations of the first k_0 principal components. Multiplying the samples by the first k right singular vectors is equivalent to orthogonally rotating the coordinate system in the k directions with the largest data variance and taking the projection of each sample in these directions as the new coordinates. Since these principal components are orthogonal to each other, the new coordinates are linearly independent in each dimension, have a diagonal covariance, and are naturally sorted in descending order of the explained variance. Subsequently, the minimum k_0 is automatically selected using the cumulative energy threshold *energy* (such as 0.65 / 0.75 / 0.85 / 0.95). Among them, the first dimension captures the largest variation in the data, the second dimension captures the second largest variation under the premise of being orthogonal to the first dimension, and so on, thereby obtaining a k -dimensional representation that is decorrelated and prioritizes information content. Σ_a also selected the k' principal component according to the above process.

$$\hat{T}_Z = Z'_{tar} * V_z^{(k)} = (U_z \Sigma_z V_z^T) * V_z^{(k)} = U_z^{(k)} * \Sigma_z^{(k)}, \quad (7)$$

$$\hat{T}_A = A'_{tar} * V_a^{(k')} = (U_a \Sigma_a V_a^T) * V_a^{(k')} = U_a^{(k')} * \Sigma_a^{(k')}. \quad (8)$$

For Z'_{tar} and A'_{tar} , the first k and k' right singular vectors are respectively taken to form $V_z^{(k)} = [v_z^1, \dots, v_z^k]$ and $V_a^{(k')} = [v_a^1, \dots, v_a^{k'}]$. Then, Z'_{tar} and A'_{tar} are right multiplied by $V_z^{(k)}$ and $V_a^{(k')}$ to obtain $\hat{T}_Z \in \mathbb{R}^{M \times k}$ and $\hat{T}_A \in \mathbb{R}^{M \times k'}$, which indicates that we have completed the positive exchange basis from the original feature coordinates to the principal component coordinates, yielding k -dimensional and k' -dimensional representations that are mutually uncorrelated, respectively. The variances of each column naturally decreasing in the order of $\sigma_1^2 \geq \dots \geq \sigma_k^2$. Thus, a low-dimensional embedding is obtained in the order of decreasing explained variance.

$$T_Z = \hat{T}_Z * (\Sigma_z^{(k)} + \varepsilon I)^{-1} = U_z^{(k)} \Sigma_z^{(k)} (\Sigma_z^{(k)} + \varepsilon I)^{-1} \approx U_z^{(k)}, \quad (9)$$

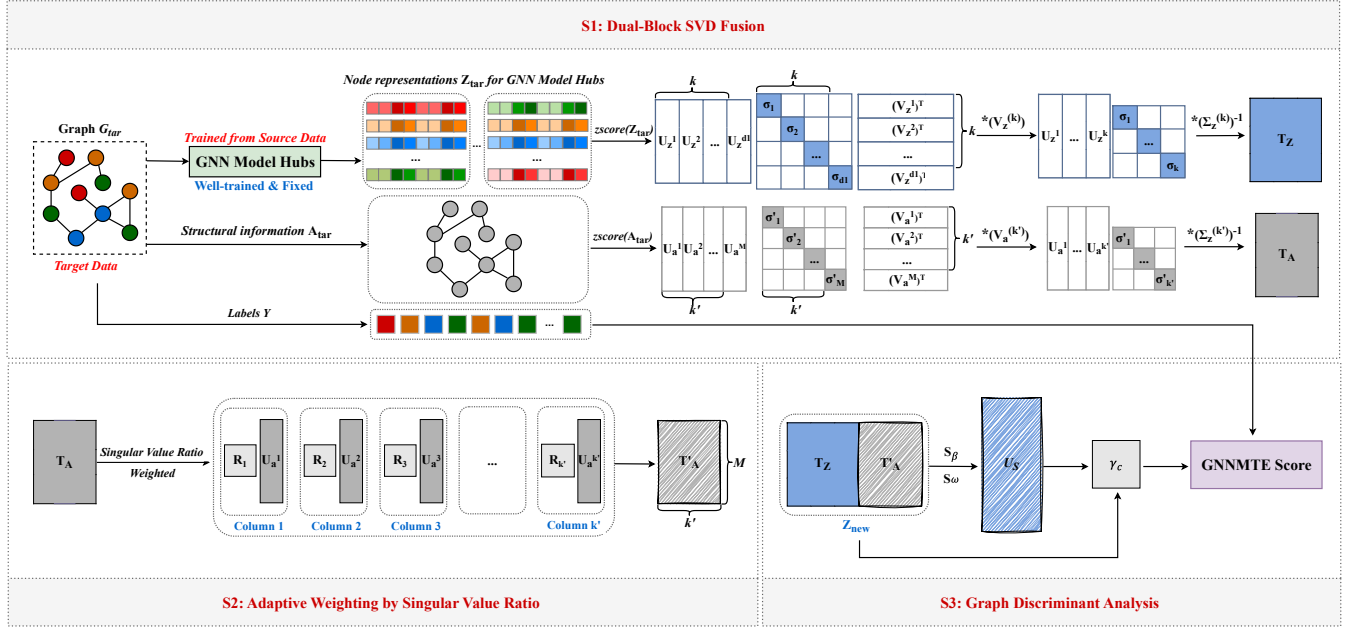


Figure 2: The overview of the proposed GNNMTE for GNN model transferability estimation, including three steps, i.e., S1: GNNMTE Dual-Block SVD Fusion; S2: Adaptive Weighting by Singular Value Ratio; S3: Graph Discriminant Analysis.

$$T_A = \hat{T}_A * (\Sigma_a^{(k')} + \varepsilon I)^{-1} = U_a^{(k')} \Sigma_a^{(k')} (\Sigma_a^{(k')} + \varepsilon I)^{-1} \approx U_a^{(k')}, \quad (10)$$

where ε is a small positive scalar, avoiding the situation where the denominator in the formula is zero. Considering that the variances in different principal component directions may vary significantly, to prevent the direction with larger variance from dominating in the distance metric and objective function, decorrelation is performed in the principal component coordinates and each component is scaled to unit variance, thereby obtaining a scale-balanced representation. Through the above process, $T_Z \in \mathbb{R}^{M \times k}$ and $T_A \in \mathbb{R}^{M \times k'}$ are obtained respectively.

3.2 Adaptive Weighting by Singular Value Ratio

In this part, we use the proportion of singular values in different columns for weighting to guide the identification and extraction of important information. By calculating the proportion of singular values in different columns, we can identify the components that have the greatest impact on the graph, thereby allocating more attention and weight to these parts.

By analyzing the proportion changes of singular values of different spectral components, Zhang et al. [44] found that after fine-tuning, the proportion of larger singular values increased, while that of smaller ones decreased. This indicates that during the fine-tuning process, the distribution of singular values tends to concentrate on the components with larger singular values, suggesting that more transferable components play a greater role in the fine-tuned features. Therefore, our method weights different columns based on the proportion of singular values of different columns, thereby enhancing the influence of the proportion of singular values on different columns. The singular value ratio adaptive column weighting

is as follows:

$$T'_A = T_A * \left(\frac{\sigma_i^2}{\sum_{j=1}^{k'} \sigma_j^2} \right)^{\frac{1}{2}}, \quad i = 1, \dots, k' \quad (11)$$

where i represents the i -th column of T_A and each column of T_A is weighted according to the proportion of singular values. Since the dual-block SVD fusion part may lead to over-averaging by flattening the variances of each column, it cannot distinguish between strong and weak directions. Therefore, by applying column weighting based on the singular value ratio, the weights of important directions are gently pulled back. The column weighting based on the singular value ratio strikes a balance between averaging and dominance. Singular value ratio column weighting is a continuously adjustable regularization between complete averaging and complete dominance. The spectral energy distribution varies among different graphs. This proportion-based weighting naturally possesses scale invariance and data adaptability, making our method more effective across different datasets.

3.3 Graph Discriminant Analysis

This section describes the implementation of the GNN model transferability evaluation score to derive a classification score based on graph discriminant analysis. It assesses separability on the target graph, converting model transferability and training value into a fast, interpretable, and rankable score for model selection. Our proposed GNNMTE finds a linear combination of features that better separate the classes of graph data. The goal of GNNMTE is to make the information of structure and attribute work together at a comparable intensity and to find the best projection direction by using these divergence matrices, thereby achieving the maximum

separation of different categories. GNNMTE aims to find the best linear combination features to maximize the separability between different classes while minimizing the intra-class scatter. When calculating the intra-class scatter matrix, the mean and covariance of each category's features are used, and ultimately a square matrix with the same number of features is obtained. Currently, the representation of the combination of new features and structural information is as follows:

$$\mathbf{Z}_{\text{new}} = \text{concatenate} [\mathbf{T}_Z, \mathbf{T}'_A], \quad (12)$$

The $\mathbf{T}_Z$ and $\mathbf{T}'_A$ obtained through the above-mentioned dual-block SVD fusion and adaptive weighting by singular value ratio processes are combined as a new combined representation integrating structural information and used as the input of graph linear discriminant analysis. We then construct the combined representation $\mathbf{Z}_{\text{new}} \in \mathbb{R}^{M \times (k+k')}$ by column-wise concatenation.

$$\mu_c = \frac{1}{M_c} \sum_{i \in C} \mathbf{Z}_{\text{new}}^{(i)}, \quad \mu = \frac{1}{M} \sum_{i=1}^M \mathbf{Z}_{\text{new}}^{(i)} \quad (13)$$

where μ_c represents the mean value of the combined feature for class c , μ represents the overall mean of all samples in $\mathbf{Z}_{\text{new}}$, M_c is the number of samples in class c , and M represents the total number of samples. These are used to calculate the inter-class and intra-class scatter matrices. let $\mathbf{S}_\beta$ and $\mathbf{S}_\omega$ denote the inter-class and intra-class scatter matrices, respectively.

$$\mathbf{S}_\beta = \sum_{c=1}^C M_c (\mu_c - \mu) (\mu_c - \mu)^T, \quad (14)$$

$$\mathbf{S}_\omega = \sum_{c=1}^C \sum_{i \in C} (\mathbf{Z}_{\text{new}}^{(i)} - \mu_c) (\mathbf{Z}_{\text{new}}^{(i)} - \mu_c)^T, \quad (15)$$

The goal is to project the features so that the inter-class variance is maximized relative to the intra-class variance. Following Ghogh et al. [10], this reduces to solving a generalized eigenvalue problem.

$$\mathbf{U}_S = \text{eig} \left((\mathbf{S}_\omega + \epsilon \mathbf{I})^{-1} \mathbf{S}_\beta \right), \quad (16)$$

where ϵ is a small positive scalar that guarantees the invertibility of the intra-class scatter matrix $\mathbf{S}_\omega$. Using Eq. (16), we obtain the eigenvector $\mathbf{U}_S$ that specify the optimal projection directions for class discrimination. The eigenvector define a new feature space in which the data are optimally separable. In our method, we assume that the features of each class follow a normal distribution centered on its projected class mean. According to Bayes' theorem, the score function for a sample labeled as c is shown in Eq. (17). Then, we calculate the GNN model transferability evaluation score S_{glda} using:

$$\gamma_c = \mathbf{Z}_{\text{new}} \mathbf{U}_S \mathbf{U}_S^T \mu_c^T - \frac{1}{2} \mu_c \mathbf{U}_S \mathbf{U}_S^T \mu_c^T + \log \left(\frac{M_c}{M} \right), \quad (17)$$

$$S_{glnmte} = \frac{1}{M} \sum_{m=1}^M \frac{e^{\gamma_y}}{\sum_{c=1}^C e^{\gamma_c}}, \quad (18)$$

where y is the ground-truth class label, M is the total number of samples, and C is the number of classes. The S_{glnmte} represents the probability of correctly classifying a sample based on the calculated discriminative scores γ_c for each class.

4 Experiments

This section is organized into six distinct parts to evaluate the proposed feature perturbation method. Section 4.1 outlines the experimental setup; Section 4.2 presents the results; Section 4.3 covers the ablation study; Section 4.4 examines hyper-parameter sensitivity; Section 4.5 analyzes time complexity; Section 4.6 shows correlation visualization.

4.1 Experimental Setup

In this part, we introduce the datasets, models, and fine-tuning models used in the experiment. Among them, the datasets include the source graph datasets and the target graph datasets. The models include the pre-trained models trained on the source graph dataset. The fine-tuning models refer to the fine-tuning of the pre-trained models on the source training set on the target graph dataset in order to obtain the 'Ground Truth'.

Datasets. In this paper, we conduct experiments on several datasets, including Cora [39], Amazon [39], ACMv9 [38], DBLPv8 [38] and Citationv2 [38] datasets. To ensure consistency of the dimensions of the dataset, several new datasets [39] are generated from the Cora and Amazon datasets. The new datasets generated from the original datasets create node labels Y and false features X_2 related to the environment ID using the original node features X_1 . Each generated graph has a unique environment ID to simulate different distribution environments. The new datasets contain ten different graphs, ensuring diversity. Due to the interconnection of nodes in the graph structure, the generated data present the characteristic of being non-independent and identically distributed, which increases the complexity of model training and evaluation. These features enable the new datasets to effectively test and evaluate the robustness and adaptability of the model in the face of distribution changes. At the same time, the same training set, validation set, and test set masks are set for the newly generated datasets respectively. In addition, we also evaluate the GNN model transferability in the following six cases, namely $A \rightarrow D$, $A \rightarrow C$, $C \rightarrow A$, $C \rightarrow D$, $D \rightarrow A$ and $D \rightarrow C$, where A , C and D represent ACMv9, DBLPv8 and Citationv2 respectively. DBLPv8, ACMv9 and Citationv2 are citation networks from different original sources (DBLP, ACM, and Microsoft Academic Graph, respectively) by following the processing of [38]. Each dataset shares the same label space with six categories of node class labels, including 'Database', 'Data mining', 'Artificial intelligent', 'Computer vision', 'Information Security', and 'High-performance computing'. The GNNs are well-trained on the ACMv9 dataset and evaluated on the Citationv2 and DBLPv8 datasets, i.e., $A \rightarrow C$ and $A \rightarrow D$, respectively. Meanwhile, the GNNs are well-trained on the Citationv2 dataset and evaluated on the ACMv9 and DBLPv8 datasets, i.e., $C \rightarrow A$ and $C \rightarrow D$, respectively.

GNN Models. We select the GCN [18], GAT [35], GraphSAGE [13], and SGFormer [40], as well as the baseline MLP [29] model that is prevalent for graph learning. We use the source datasets to pre-train the selected models, obtaining the pre-trained GCN, GAT, GraphSAGE, MLP and SGFormer models.

Finetune Models. To address the issue of model transferability estimation, in order to establish a reliable basis for comparing pre-trained models, each model is first fine-tuned on the target dataset,

Table 1: GNN model transferability evaluation method GNNMTE and its comparative methods with transferability evaluation scores for different models and the ranking of the transferability of different models under the current method. GNNs are well-trained on the Citationv2 dataset and evaluated on the ACMv9 dataset.

Citationv2→ACMv9	GCN	$rank_{GCN}$	GAT	$rank_{GAT}$	SAGE	$rank_{SAGE}$	MLP	$rank_{MLP}$	SGFormer	$rank_{SGF}$
NLEP	0.481	1	0.440	3	0.440	3	0.358	5	0.473	2
LogME	-0.085	1	-0.150	4	-0.113	3	-0.177	5	-0.096	2
SFDA	0.448	1	0.381	4	0.425	2	0.357	5	0.420	3
GBC	-21.364	2	-23.826	4	-21.952	3	-24.725	5	-11.082	1
NCTI	0.438	1	0.383	4	0.413	2	0.336	5	0.402	3
LDA	0.326	2	0.290	4	0.318	3	0.288	5	0.373	1
Ground Truth(%)	84.620	1	82.590	2	79.890	4	72.270	5	82.190	3
GNNMTE (ours)	70.467	1	70.451	2	70.446	4	70.425	5	70.448	3

Table 2: Performance comparison (the weighted Kendall τ_w) for fine-tuning on GNN models. In each column, the best results are highlighted in bold and a larger τ_w represents a better prediction rank. Remarkably, GNNMTE achieves the highest overall average weighted Kendall τ_w score.

Method	Cora	Amazon	D→A	C→A	A→D	C→D	A→C	D→C	Average
NLEP	-0.568	0.214	0.442	0.681	0.197	0.818	0.088	-0.088	0.223
LogME	0.201	0.015	0.657	0.681	0.453	0.197	0.270	-0.007	0.308
SFDA	0.000	0.015	0.657	0.526	0.453	0.453	0.270	-0.007	0.296
GBC	0.000	0.015	0.578	0.316	-0.369	0.115	0.270	0.270	0.149
NCTI	0.000	-0.141	0.657	0.526	0.270	0.818	0.015	0.168	0.289
LDA	0.354	-0.162	0.770	0.316	-0.186	0.270	0.270	0.354	0.248
GNNMTE (ours)	0.297	0.214	0.672	1.000	0.573	1.000	0.818	0.573	0.643

and a comprehensive exploration of different hyperparameters, including learning rate and weight decay, is conducted. It is necessary to fine-tune the pre-trained model on the source dataset to adapt its performance on the target dataset. The test accuracy obtained through fine-tuning is used as the 'true label' for ranking these models. Suppose that there are L models and that the fine-tuning performance of the i -th model is denoted as G_i , which serves as the benchmark for ranking models through transferability estimation metrics.

Benchmark methods. Currently, various benchmark methods have been identified to evaluate the transferability of GNN models. These methods are applied to transferability assessment in models from fields such as computer vision and natural language processing. Previous research in the image domain can be categorized into label-based methods and source embedding-based methods. The Log-Euclidean Expectation Predictor (LEEP) [24] and Negative Conditional Entropy (NCE) [34] are two transferability estimation metrics that rely on information from the sets of source and target labels. By processing target data through the feature extractor of a pre-trained model, target embeddings can be obtained for use in source embedding methods. These benchmark methods include the NLEP [20], LogME [43], SFDA [31], NCTI [37], GBC [26], and LDA [16] methods.

4.2 Experimental Results

Overview. In this paper, we evaluate the transferability of GNN models, aiming to directly apply pre-trained GNN models from the

source graph datasets to the target graph datasets using the GNN transferability evaluation metric without fine-tuning. In the Cora dataset, 1-gat.pkl is taken as the source dataset, and the remaining nine datasets are used as target datasets. In the Amazon dataset, 1-sgc.pkl is taken as the source dataset, and the remaining nine datasets are used as target datasets. For both the Cora and Amazon datasets, a set of GNN Models Hub will be pre-trained for each source dataset. Each pre-trained GNN model hub contains five pre-trained models: GCN, GAT, GraphSAGE, MLP, and SGFormer. It is worth noting that, for both the Cora and Amazon datasets, whether calculating the accuracy of the fine-tuned test data or evaluating the transferability of the GNN models, the average value of the nine datasets is calculated. In the ACM, DBLP, and Network datasets, one dataset is selected as the source dataset, and the remaining two datasets are used as target datasets. Thus, there will be 3 sets of pre-trained GNN Models Hub in this process. A total of 25 pre-trained models are used in the experiments in this paper.

Ground truth and correlation measurement. We follow the grid search method described in [31], selecting the learning rates from $\{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$ and weight decay parameters from $\{0, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}\}$. The GNN models pre-trained on the source graph dataset are fine-tuned on the target graph dataset. By selecting appropriate learning rates and weight decay parameters, we obtain the test accuracy of the fine-tuned GNN models on different target datasets as 'Ground Truth'.

Performance comparison. GNN model transferability estimation score is obtained when the model trained by the source graph

dataset is applied to the target graph dataset and the order in which different methods select models for the current target graph dataset is presented in Table 8. The GNN model transferability estimation scores of our method GNNMTE and six other comparison methods, namely *NLEEP*, *logme*, *SFDA*, *GBC*, *NCTI*, and *LDA*, are all presented and our method, along with the comparison methods, is evaluated against the 'Ground Truth'. It shows that our calculated GNN model transferability estimation scores can more accurately predict the ranking of models suitable for the target graph dataset. We report the experimental results and show the ranking performance of the GNN models in Table 2. As can be seen from the table, GNNMTE method demonstrates notable improvements in GNN transferability estimation across all previous metrics. Our method improves the average results of the entire dataset by 42.0%, 33.5%, 34.7%, 49.4%, 35.4% and 39.5% respectively compared to the comparison methods *NLEEP*, *LogMe*, *SFDA*, *GBC*, *NCTI* and *LDA*, and with the *GBC* metric benefiting the most from this approach. The table reports an average improvement of 39.1% on previous transferability estimation methods, indicating the effectiveness of our approach. Especially on the Citationv2→ACMv9, ACMv9→DBLPv8, Citationv2→DBLPv8, ACMv9→Citationv2 and DBLPv8→Citationv2 cases, our method demonstrated significant competitiveness and the weighted Kendall coefficient calculated with the 'Ground Truth' reach correlations of 1.000, 0.573, 1.000, 0.818, and 0.573, respectively. In particular, our proposed method achieves a considerable improvement on the average value across all graph datasets reaches a maximum of 0.643, outperforming other existing methods and verifying the effectiveness of our proposed method. Therefore, our method can more accurately select the most suitable GNN model for different target graph datasets without fine-tuning.

4.3 Ablation Study

In this part, to verify the effectiveness of each component in the proposed method, we design different baseline transferability estimation metrics for different important components to conduct ablation studies. The ablation experiments we design include 'w/o SVD', 'w/o SVD(A'_{tar})', 'w/o T/Σ ' and 'w/o Weight'. 'w/o SVD' indicates that SVD is not performed on Z'_{tar} and A'_{tar} , and they are directly concatenated and input into graph linear discriminant analysis. 'w/o SVD(A'_{tar})' indicates that SVD is only performed on Z'_{tar} , but not on A'_{tar} . 'w/o T/Σ ' indicates that the operations in Eq. (9) and Eq. (10) are not performed on T by dividing by $\Sigma + \epsilon I$. 'w/o Weight' indicates that the singular value weighting operation by column is not performed on T_A . The comparison results of the ablation experiments are shown in Table 3. Our method consistently outperforms all ablation methods in systematic ablation studies. On average of all datasets, it improves by an average of 21.1% compared to the four variants with key components removed, highlighting the crucial contribution of our design to overall performance.

4.4 Hyper-parameter Sensitivity Analysis

In this section, we evaluate the impact of a key hyperparameter, namely *energy*, which represents the explained cumulative variance ratio and determines the number of principal components retained. As shown in Fig. 3, the optimal values *energy* for different

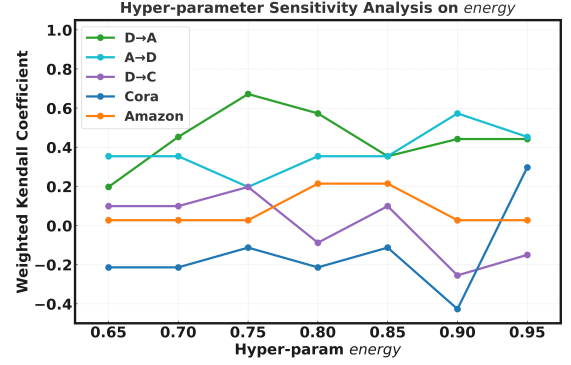


Figure 3: Hyper-parameter Sensitivity Analysis on *energy*.

datasets vary significantly. Specifically, the optimal performance is achieved at *energy* value of 0.75, 0.90, 0.75, 0.95 and 0.80 for the DBLPv8→ACMv9, ACMv9→DBLPv8, DBLPv8→Citationv2, Cora and Amazon, respectively. Each dataset finds its own optimal balance, offering the best performance and making it ideal for practical use.

4.5 Time Complexity

We analyze the time complexity of the proposed GNNMTE method. For S1 dual-block SVD fusion, SVD is performed on Z'_{tar} and A'_{tar} respectively, with the retained dimensions being k and k' , and the time complexities being $O(Md_1k + (M + d_1)k^2)$ and $O(M^2k' + M(k')^2)$ respectively. Then, the time complexities of right multiplying by Σ_z^{-1} and Σ_a^{-1} are $O(Mk)$ and $O(Mk')$ respectively. For S2 adaptive weighting by singular value ratio, the weight calculation of the singular value ratio weighting itself is $O(k + k')$, and if applied to each row, it becomes $O(M(k + k'))$. Let $D = k + k'$. The singular value ratio weighting costs $O(MD)$. For S3 graph discriminant analysis, computing the class means and scatter matrices costs $O(MD) + O(MD^2)$, and solving the $D \times D$ generalized eigenvalue problem costs $O(D^3)$. The scoring stage costs $O((M + C)Dr + MCr)$ with $r \leq D$. Since D is smaller than M , the overall time complexity can be summarized as $O(Md_1k) + O(M^2k')$.

4.6 Correlation Visualization

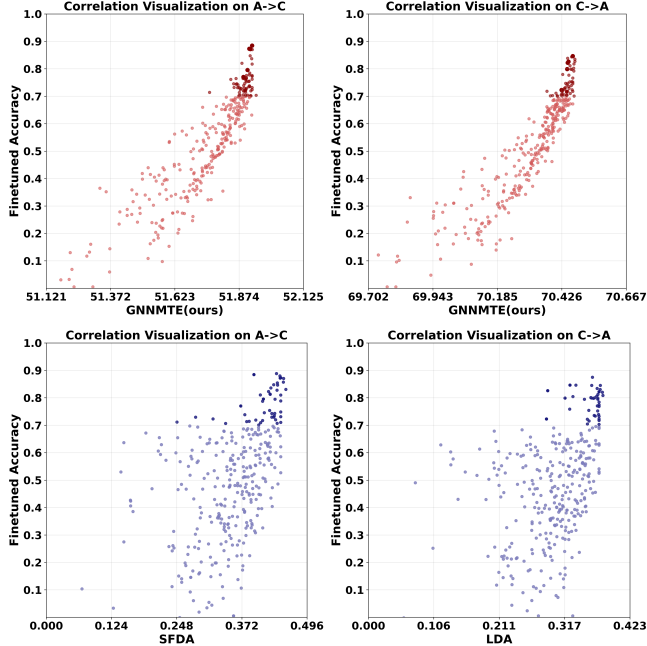
We visualize the correlation between the fine-tuned accuracy rates as ground-truth under different datasets and the GNNMTE we proposed, and compared them with the comparison methods on ACMv9→Citationv2 and Citationv2→ACMv9 cases in Fig. 4. It can be observed that the GNNMTE we proposed has a stronger correlation compared to other baselines, verifying its effectiveness associated with the fine-tuned accuracy as the ground-truth.

5 Related Works

In recent years, many researchers have focused on model transfer learning and have proposed various evaluation methods. Currently, both domestically and internationally, transferability estimation techniques have been developed for fields such as image recognition, natural language processing, speech recognition, recommendation systems, and reinforcement learning. Their model transferability estimation methods can be divided into two major categories: source-free model transferability estimation (SF-MTE) and

Table 3: Ablation study performance comparison (the weighted Kendall τ_w) for fine-tuning on GNN models. In each column, the best results are highlighted in bold.

Method	Cora	Amazon	D→A	C→A	A→D	C→D	A→C	D→C	Average
w/o SVD	0.088	0.115	0.770	0.681	0.453	0.872	0.453	0.197	0.453
w/o SVD(A'_{tar})	-0.113	0.214	0.354	1.000	0.354	0.901	0.672	0.099	0.435
w/o T/ Σ	-0.113	0.027	0.901	0.316	0.270	0.818	0.672	0.354	0.406
w/o Weight	-0.113	0.214	0.354	1.000	0.354	0.901	0.672	0.099	0.435
GNNMTE (ours)	0.297	0.214	0.672	1.000	0.573	1.000	0.818	0.573	0.643

**Figure 4: Correlation visualization for our GNNMTE on A→C, our GNNMTE on C→A, SFDA on A→C and LDA on C→A.**

source-dependent model transferability estimation (SD-MTE) [3]. The source-free model transferability estimation methods can be further classified into static methods and dynamic methods. The static methods for estimating the transferability of source-free models mainly include methods based on feature structure [1, 26, 42], Bayesian statistics [20, 24, 31], information theory [15, 34], and matrix analysis [1, 9]. Dynamic methods for assessing the transferability of source-free models include energy models [11, 19], linear frameworks [17, 30, 43], and model vectorization [4, 45]. The source-dependent model transferability estimation methods can also be divided into static methods and dynamic methods. Since the source data are available, the static methods for source-dependent model transferability estimation generally include distribution matching [22, 33], dual graphs [5, 6]. Dynamic methods for assessing the transferability of source-dependent models include methods based on gradient information [27, 32], representation space transformation [41], and linear frameworks [12, 25]. Among them, some of these model transferability estimation methods are representative and widely applied, each having their own unique characteristics. The GBC metric [26] assesses the degree of overlap

between classes in the embedded space of target data, modeling the features of each class as a Gaussian distribution and using the Bhattacharyya coefficient to measure the inter-class separation between class pairs. A higher GBC value indicates a greater degree of overlap. LEEP [24] and its variants calculate the average log-likelihood of the log-expectation empirical predictor, which is a non-parametric classifier based on the joint distribution of source and target distributions. NLEEP [20] improves LEEP by replacing the output layer of the model with a Gaussian mixture model [28]. LogME [43] fits a linear model on target features and estimates the maximum log-evidence under a Gaussian noise assumption to measure how well pre-trained features match target labels, thereby evaluating transferability. SFDA [31] employs Fisher Discriminant Analysis (FDA) [23] and introduces ConfMix. The principle of FDA is to learn a linear transformation that maximizes inter-class separation via within-class and between-class scatter. ConfMix then generates challenging negative samples. The NCTI [37] metric quantifies the gap between a model’s embedding and the ideal neural collapse embedding. LDA [16] maximizes the separation between different classes by finding the best linear combination of features.

Despite the lack of studies on GNN model transferability estimation, the closely related problem of GNN model performance evaluation has been actively investigated, including GNNEvaluator [48] and LEBED [47], respectively construct augmented subgraphs and employ iterative retraining on unlabeled test graphs to assess model robustness and prediction reliability.

6 Conclusions and Discussions

In this paper, we study a new problem, GNN model transferability estimation, to understand and evaluate the performance of well-trained GNNs in terms of their transferability to target graph datasets. We propose a novel method, called GNNMTE, which does not require fine-tuning on target graph datasets but directly calculates the GNNMTE score to evaluate the transferability of GNN models. Extensive experiments and analyses verify the effectiveness of our proposed method. In the future, we will continue to explore more methods for this innovative problem and will also consider the evaluation of the GNN model transferability among data in different dimensions. We hope that our research may provide useful experience and advance the study of graphs and GNN models.

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A More Experimental Results

Table 4: The test accuracy (%) on the target datasets obtained through fine-tuning is used as the 'ground truth' for GNN model transferability estimation.

Dataset	GCN	GAT	SAGE	MLP	SGFormer
Cora	70.900	66.520	69.110	47.380	70.080
Amazon	90.940	89.160	86.370	77.340	88.260
ACMv9→DBLPv8	98.570	98.480	97.760	69.920	95.340
ACMv9→Citationv2	87.180	88.450	79.560	77.010	87.290
DBLPv8→ACMv9	83.000	82.730	81.310	74.630	83.670
DBLPv8→Citationv2	86.330	88.240	83.580	76.800	87.080
Citationv2→ACMv9	84.620	82.590	79.890	72.270	82.190
Citationv2→DBLPv8	98.480	98.570	97.760	69.470	95.790

Table 5: GNN model transferability evaluation method GNNMTE and its comparative methods with transferability evaluation scores for different models on Cora and Amazon. GNNs are well-trained on the source datasets.

Method	Cora					Amazon				
	GCN	GAT	SAGE	MLP	SGFormer	GCN	GAT	SAGE	MLP	SGFormer
NLEP	0.542	0.545	0.515	0.659	0.489	0.586	0.740	0.643	0.583	0.568
LogME	0.210	0.127	0.203	0.306	1.116	0.884	0.921	0.908	0.868	1.331
SFDA	0.552	0.472	0.568	0.733	0.987	0.728	0.762	0.759	0.718	0.966
gbc	-49.373	-51.986	-43.548	-20.873	-3.367	-46.387	-31.367	-35.187	-61.105	-13.733
NCTI	0.551	0.464	0.556	0.612	0.907	0.727	0.751	0.757	0.712	0.914
LDA	0.275	0.202	0.279	0.206	0.419	0.453	0.596	0.555	0.494	0.459
GNNMTE (ours)	-15.465	-15.522	-15.536	-15.528	-15.595	404.265	404.309	404.285	404.265	404.259

In this section, We present more experimental results, including the ground truth after fine-tuning on the target graph datasets and the GNNMTE scores obtained on different methods and different datasets.

Pretraining on the source graph dataset. We pretrain the GNN in an end-to-end manner on the source dataset, leveraging the complete graph information provided by the source domain, including nodes, edges, and node features, together with the predefined train/validation/test split masks. Specifically, for each source dataset, we train different GNN models separately. At each epoch, the model minimizes the supervised classification loss on the labeled nodes in the training split, while the validation loss and accuracy are computed on the validation split for model selection and learning-rate scheduling. We further adopt early stopping: training is terminated once the validation loss no longer improves, and the checkpoint achieving the lowest validation loss is saved as the pre-trained model for the corresponding source-backbone pair.

Fine-tuning on the target graph dataset. Table 4 reports the target-domain ground truth obtained by fine-tuning the pretrained GNN model on the target dataset. To derive this ground truth criterion, we initialize the model with the parameters pretrained on the source graph and conduct supervised fine-tuning on the target graph. During fine-tuning, the model takes the target graph structure and node features as input and is optimized using only the labeled nodes in the target training split. Early stopping and checkpoint selection are performed based on the target validation split to mitigate overfitting and ensure a fair comparison. Finally, we treat

the test accuracy achieved on the target test split after fine-tuning as the target-domain ground truth and use it as 'Ground Truth' for evaluating GNN transferability.

GNNMTE score on the target training dataset. We obtain GNNMTE scores under eight cases: Cora, Amazon, $A \rightarrow D$, $A \rightarrow C$, $D \rightarrow A$, $D \rightarrow C$, $C \rightarrow A$, and $C \rightarrow D$. Results for the Cora and Amazon cases are presented in Table 5; the $A \rightarrow D$ and $A \rightarrow C$ cases are presented in Table 6; the $D \rightarrow A$ and $D \rightarrow C$ cases are presented in Table 7; and the $C \rightarrow A$ and $C \rightarrow D$ cases are presented in Table 8. For each case, we compute transferability scores for all candidate GNN model hubs under different methods, yielding a complete set of model transferability estimation scores. We then rank the five GNN models according to their scores in each case, and select the best-performing GNN model for the corresponding target graph dataset case based on this ranking. Finally, we compare the scores of the five GNN models on the target graph dataset against the 'Ground Truth', and use the weighted Kendall correlation coefficient to evaluate the effectiveness of the proposed method.

Environment. Our experimental environment is equipped with an NVIDIA GeForce RTX 4090 graphics card, driver version 575.51.03, CUDA version 12.9, and is powered by a 12th Gen Intel(R) Core(TM) i9-12900K processor.

Code implementation. We release the implementation of GNNMTE at <https://github.com/Huanchang-Ma/GNNMTE/>. Our results reported in this paper are reproducible using the provided code and configurations.

Table 6: GNN model transferability evaluation method GNNMTE and its comparative methods with transferability evaluation scores for different models. GNNs are well-trained on the ACMv9 dataset and evaluated on the DBLPv8 and Citationv2 datasets.

Method	ACMv9→DBLPv8					ACMv9→Citationv2				
	GCN	GAT	SAGE	MLP	SGFormer	GCN	GAT	SAGE	MLP	SGFormer
$\mathcal{N}$ LLEP	0.464	0.551	0.562	0.314	0.360	0.507	0.474	0.482	0.420	0.457
LogME	-0.205	-0.155	-0.192	-0.301	-0.210	-0.114	-0.156	-0.135	-0.162	-0.100
SFDA	0.370	0.468	0.460	0.289	0.364	0.445	0.395	0.413	0.370	0.446
gbc	-21.620	-18.243	-18.044	-24.064	-16.489	-21.839	-23.754	-22.405	-24.231	-12.785
NCTI	0.337	0.457	0.414	0.265	0.338	0.436	0.399	0.403	0.349	0.420
LDA	0.218	0.246	0.229	0.195	0.265	0.324	0.288	0.313	0.281	0.407
GNNMTE (ours)	-25.112	-25.098	-25.120	-25.156	-25.167	51.918	51.925	51.907	51.891	51.914

Table 7: GNN model transferability evaluation method GNNMTE and its comparative methods with transferability evaluation scores for different models. GNNs are well-trained on the DBLPv8 dataset and evaluated on the ACMv9 and Citationv2 datasets.

Method	DBLPv8→ACMv9					DBLPv8→Citationv2				
	GCN	GAT	SAGE	MLP	SGFormer	GCN	GAT	SAGE	MLP	SGFormer
$\mathcal{N}$ LLEP	0.571	0.446	0.397	0.458	0.508	0.558	0.425	0.400	0.440	0.500
LogME	-0.073	-0.145	-0.139	-0.112	-0.024	-0.125	-0.187	-0.165	-0.146	-0.066
SFDA	0.471	0.365	0.400	0.402	0.506	0.445	0.343	0.380	0.384	0.480
gbc	-23.808	-25.427	-23.574	-25.298	-7.331	-24.992	-25.568	-25.173	-26.002	-8.815
NCTI	0.471	0.365	0.366	0.389	0.487	0.445	0.350	0.347	0.362	0.450
LDA	0.426	0.272	0.304	0.275	0.463	0.383	0.257	0.253	0.255	0.447
GNNMTE (ours)	70.459	70.449	70.429	70.423	70.450	51.908	51.919	51.882	51.889	51.928

Table 8: GNN model transferability evaluation method GNNMTE and its comparative methods with transferability evaluation scores for different models. GNNs are well-trained on the Citationv2 dataset and evaluated on the ACMv9 and DBLPv8 datasets.

Method	Citationv2→ACMv9					Citationv2→DBLPv8				
	GCN	GAT	SAGE	MLP	SGFormer	GCN	GAT	SAGE	MLP	SGFormer
$\mathcal{N}$ LLEP	0.481	0.440	0.440	0.358	0.473	0.454	0.501	0.468	0.278	0.348
LogME	-0.085	-0.150	-0.113	-0.177	-0.096	-0.223	-0.228	-0.219	-0.321	-0.244
SFDA	0.448	0.381	0.425	0.357	0.420	0.350	0.371	0.397	0.267	0.337
gbc	-21.364	-23.826	-21.952	-24.725	-11.082	-21.392	-19.883	-20.141	-23.889	-17.206
NCTI	0.438	0.383	0.413	0.336	0.402	0.322	0.365	0.353	0.253	0.313
LDA	0.326	0.290	0.318	0.288	0.373	0.209	0.231	0.205	0.198	0.243
GNNMTE (ours)	70.467	70.451	70.446	70.425	70.448	-25.121	-25.110	-25.135	-25.163	-25.149